Project Plan: Speech Commands Classification with Transformers

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Abstract

This document presents a project plan for a deep learning study on speech command classification with Transformer-based models. The goal is to compare several architectures on the Speech Commands dataset, including a lightweight convolutional baseline, a task-specific Transformer, and a pretrained Transformer-based model. Particular attention is given to the `silence` and `unknown` classes, which are important for robust keyword spotting. The project includes a comparison of architectures, an investigation of selected parameter changes, and an evaluation based on standard classification metrics and confusion matrices.

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1 Introduction

Speech command classification, also known as keyword spotting, is the task of assigning a short audio recording to one of several predefined spoken commands. It is used in voice interfaces, embedded assistants, and low-latency control systems. Although the problem is smaller than full speech recognition, it remains challenging because of background noise, speaker variability, and the need to reject non-target inputs.

The aim of this project is to investigate Transformer-based approaches for speech command classification. The study will compare Transformers with a simple convolutional baseline and will focus especially on the recognition of the `silence` and `unknown` classes.

2 Research Problem and Objectives

The main research question is:

How effective are Transformer-based models for 12-class speech command classification, and can dedicated handling of `silence` and `unknown` improve robustness?

The main objectives are:

1. to build a reproducible speech command classification pipeline,
2. to compare multiple neural architectures, including at least one Transformer,
3. to investigate the influence of selected parameters,
4. to analyze classification errors using confusion matrices,
5. to study methods for better recognition of `silence` and `unknown`.

3 Dataset and Task Setup

The project will use the *Speech Commands* dataset in the standard 12-class setup. The target labels are:

`yes, no, up, down, left, right, on, off, stop, go`

together with two additional classes:

- `silence`,
- `unknown`.

This formulation is well suited to the project because it includes both command recognition and rejection of irrelevant inputs. The study will therefore address not only accuracy on target commands but also robustness in non-command cases.

4 Theoretical Motivation

Convolutional neural networks are a natural baseline for keyword spotting because spectrograms can be treated as image-like inputs and CNNs are efficient and stable. However, Transformer models have become attractive because self-attention can capture temporal relationships more flexibly than purely convolutional approaches.

In this project, Transformers are particularly relevant for two reasons. First, lightweight task-specific Transformer architectures have been proposed for keyword spotting. Second, pretrained audio Transformers allow transfer learning and may achieve strong performance after fine-tuning.

The planned literature background includes the original Transformer model, Keyword Transformer, Audio Spectrogram Transformer, and standard methods for speech feature extraction and augmentation.

5 Input Representation and Preprocessing

The planned input representation is a **log-Mel spectrogram**. This representation is standard in speech classification and is compatible with both CNN and Transformer-based models.

The preprocessing pipeline will include:

- fixed audio sampling rate,
- padding or trimming recordings to a fixed duration,
- conversion to log-Mel spectrograms,
- normalization of input features.

The planned augmentations are:

- time shifting,
- background noise injection,
- SpecAugment.

These steps are expected to improve generalization and robustness.

6 Planned Models

The project will compare three model families.

6.1 CNN Baseline

The first model will be a small convolutional neural network operating on log-Mel spectrograms. It will provide a simple and efficient baseline for comparison with more advanced models.

6.2 Task-Specific Transformer

The main research model will be a lightweight Transformer inspired by keyword spotting architectures such as Keyword Transformer. It will process spectrogram-derived tokens and use self-attention to model temporal dependencies in the audio signal.

6.3 Pretrained Audio Transformer

The third model will be a pretrained Transformer-based architecture, preferably Audio Spectrogram Transformer (AST), fine-tuned for speech command classification. This model will represent the transfer learning setting and will allow the study to compare training from scratch with pretrained representations.

7 Handling of `silence` and `unknown`

One of the main goals of the project is to improve recognition of the `silence` and `unknown` classes. For this reason, three strategies will be compared.

7.1 Flat 12-Class Classification

The first approach is a standard single-stage classifier that directly predicts one of the 12 classes.

7.2 Flat Classification with Rebalancing

The second approach keeps the same 12-class structure but modifies the training procedure using class weighting, oversampling, or a similar balancing method. The objective is to improve recognition of difficult classes.

7.3 Hierarchical Pipeline

The third approach is a hierarchical pipeline. In the first stage, the model predicts whether the input belongs to:

- `silence`,
- `target-command`,
- `unknown-or-other`.

If the input is classified as a target command, a second-stage classifier predicts one of the ten command labels. Otherwise, the output is mapped directly to `silence` or `unknown`. This design may improve rejection of non-command inputs and will serve as the main original element of the project.

8 Planned Experiments

The experiments will be organized into three groups.

8.1 Architecture Comparison

The first group will compare:

- CNN baseline,
- task-specific Transformer,
- pretrained audio Transformer.

8.2 Parameter Influence

The second group will study selected parameter changes, especially for the Transformer model. The planned comparisons include:

- smaller vs. larger Transformer configuration,
- training with and without selected augmentations.

8.3 Difficult-Class Handling

The third group will compare the three strategies for `silence` and `unknown`:

- flat 12-class classification,
- flat classification with rebalancing,
- hierarchical pipeline.

9 Evaluation Protocol

The main evaluation metric will be classification accuracy. However, because the task includes difficult rejection classes, additional measures will also be used.

The planned evaluation will include:

- overall accuracy,
- macro F1-score,
- per-class recall for `silence` and `unknown`,
- confusion matrix.

The confusion matrix will be particularly important in the final analysis because it will show whether the models confuse target commands with non-command inputs.

10 Expected Contributions and Conclusion

The expected contributions of the project are:

1. a reproducible speech command classification pipeline,
2. a comparison between CNN and Transformer-based approaches,
3. an analysis of selected model and training choices,
4. a focused study of `silence` and `unknown`,
5. a hierarchical classification strategy as a more original project component.

In summary, the project is designed as a focused study on Transformer-based keyword spotting with particular attention to difficult rejection classes. It satisfies the formal requirements of the assignment while maintaining a clear and manageable experimental structure.

References

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